Quantum Machine Learning for Terrestrial Radar Vessel Type Classification: Does QML Provide the Edge?

Research Question 3: Does Quantum ML offer measurable advantages over classical approaches?

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Abstract

This paper investigates whether Quantum Machine Learning (QML) approaches — specifically the Variational Quantum Classifier (VQC) and Hybrid Quantum–Classical Neural Network (HQNN) — offer measurable classification advantages over classical models (Random Forest, SVM, MLP, Gradient Boosting Trees) for radar-based vessel type classification from a coastal X-band surveillance radar. Evaluating on the same 5-feature corrected dataset (100,430 observations, 5 vessel types), we find that HQNN achieves 82.3% accuracy and F1 Macro 82.8% (AUC 0.953), competitive with Random Forest (89.6%) and approaching Gradient Boosting (90.0%), while outperforming SVM (80.8%) and MLP (77.9%). VQC achieves 72.4% (AUC 0.901), limited by circuit expressibility at 5 qubits. A soft-vote ensemble (VQC+HQNN+GBT) achieves 84.2% (AUC 0.979). Training is performed on a classical quantum simulator (`lightning.qubit`) using the adjoint differentiation method with a PyTorch optimisation backend. We analyse the theoretical conditions for quantum advantage and discuss implications for operational maritime domain awareness systems.

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1 Introduction

The potential advantage of quantum computing for machine learning has been a subject of intense research since the proposal of quantum speedups for linear algebra [5] and kernel methods [1]. In practice, demonstrating a clear quantum advantage on real-world classification problems remains an open challenge, particularly for near-term noisy intermediate-scale quantum (NISQ) devices where circuit depth is limited.

This paper addresses **RQ3**: *Does Quantum Machine Learning offer measurable advantages over classical approaches in accuracy, generalisation, or confidence calibration for the radar vessel classification problem?*

We approach this systematically:

1. Implement three QML architectures (VQC, HQNN, QKE) on a classical quantum simulator.
2. Train and evaluate each QML model on the same dataset and features used for classical ML in Paper 2.
3. Compare all models on a comprehensive set of metrics.
4. Analyse confidence score distributions to assess whether QML produces better-calibrated uncertainty estimates.
5. Discuss the theoretical basis for any observed differences.

2 Quantum Computing Background

2.1 Qubits and Quantum Gates

A quantum bit (qubit) is a two-level quantum system with state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \tag{1}$$

An n-qubit system occupies a 2^n-dimensional Hilbert space, allowing exponentially many basis states to be superposed. Quantum gates are unitary transformations $U \in \mathbb{C}^{2^n \times 2^n}$, $UU^\dagger = I$.

Common single-qubit rotation gates:

$$R_x(\theta) = \exp(-i\theta X/2) = \cos\tfrac{\theta}{2}I - i\sin\tfrac{\theta}{2}X \tag{2}$$
$$R_y(\theta) = \exp(-i\theta Y/2) = \cos\tfrac{\theta}{2}I - i\sin\tfrac{\theta}{2}Y \tag{3}$$
$$R_z(\theta) = \exp(-i\theta Z/2) = \cos\tfrac{\theta}{2}I - i\sin\tfrac{\theta}{2}Z \tag{4}$$

The CNOT gate entangles two qubits:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{5}$$

2.2 Quantum Measurement

The expectation value of the Pauli-Z operator on qubit i is:

$$\langle Z_i \rangle = \langle \psi | Z_i | \psi \rangle \in [-1, 1] \tag{6}$$

This maps the quantum state to a real number used as model output.

2.3 Data Encoding: Angle Embedding

Classical features $\boldsymbol{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d$ are encoded into an n-qubit state via rotation gates:

$$U_\phi(\boldsymbol{x}) = \bigotimes_{i=1}^{d} R_y(\pi x_i') \tag{7}$$

where x_i' is the normalised feature value in $[0, 1]$ and $d = n$ (one feature per qubit). After encoding:

$$R_y(\pi x_i')|0\rangle = \cos\tfrac{\pi x_i'}{2}|0\rangle + \sin\tfrac{\pi x_i'}{2}|1\rangle \tag{8}$$

Angle encoding is efficient (depth $O(n)$) but limited to n features. For $d > n$ features, either additional encoding layers or a classical pre-processing step (as in HQNN) is used.

3 QML Model Architectures

3.1 Variational Quantum Classifier (VQC)

The VQC [2] is a parameterised quantum circuit trained end-to-end via gradient descent on the classification loss.

3.1.1 Architecture

1. **Encoding layer** $U_\phi(\boldsymbol{x})$: angle encoding of 5 features into 5 qubits.

2. **Variational layers** $U_\theta(\boldsymbol{\theta})$: $L = 3$ repetitions of the Strongly Entangling Layer

(SEL) ansatz:

$$\text{SEL}_l = \text{CNOT\_ring} \cdot \prod_{i=1}^{5} R_z(\theta_{l,i,1}) R_y(\theta_{l,i,2}) R_z(\theta_{l,i,3}) \tag{9}$$

where CNOT_ring applies CNOT gates in a circular pattern ($1 \to 2, 2 \to 3, 3 \to 4, 4 \to 5, 5 \to 1$), creating full entanglement.

3. **Measurement**: Pauli-Z expectation on all 5 qubits $[\langle Z_1 \rangle, \dots, \langle Z_5 \rangle]$.

4. **Classical output layer**: Linear transformation to 5 class logits, followed by softmax.

3.1.2 Parameter Count

L layers $\times$ n qubits $\times$ 3 rotation angles $= 3 \times 5 \times 3 = 45$ variational parameters, plus the output layer ($5 \times 5 + 5 = 30$ classical parameters). Total: 75 trainable parameters.

3.1.3 Training

- Optimiser: Adam, $\eta = 0.005$, $\beta_1 = 0.9$, $\beta_2 = 0.999$
- Loss: weighted cross-entropy
- Max epochs: 150, early stopping (patience = 20)
- Gradient computation: parameter-shift rule (exact, no finite differences)

The parameter-shift rule computes exact gradients:

$$\frac{\partial}{\partial \theta_k} \langle O \rangle(\boldsymbol{\theta}) = \frac{1}{2} \left[\langle O \rangle(\boldsymbol{\theta} + \tfrac{\pi}{2} \boldsymbol{e}_k) - \langle O \rangle(\boldsymbol{\theta} - \tfrac{\pi}{2} \boldsymbol{e}_k) \right] \tag{10}$$

3.2 Hybrid Quantum–Classical Neural Network (HQNN)

The HQNN [3] embeds a quantum circuit within a classical neural network, enabling the classical layers to learn a better feature representation for the quantum circuit and to post-process quantum measurements.

3.2.1 Architecture

$$\hat{\boldsymbol{y}} = \text{softmax}(f_{\text{post}}(Q_\theta(f_{\text{pre}}(\boldsymbol{x})))) \tag{11}$$

1. **Classical pre-processing**: 2 fully connected layers (ReLU, width 32) that transform the 5 input features before quantum encoding. Output: 5 values (one per qubit).

2. **Quantum layer**: $n = 5$ qubits, $L = 2$ SEL layers, angle encoding.

3. **Classical post-processing**: Linear layer mapping 5 qubit measurements to 5 class logits.

3.2.2 Motivation for Hybrid Architecture

The classical pre-layers allow the model to:

- Learn non-linear feature combinations before quantum encoding.
- Scale features to the $[0, \pi]$ encoding range adaptively.
- Use more expressive representations than raw angle encoding allows.

This hybrid design is specifically beneficial when the raw features are not directly suitable for angle encoding (e.g., when feature scales vary significantly between variables).

3.2.3 Training

Same as VQC. Early stopping typically triggers later (around epoch 20–30) compared to VQC (epoch 8–32), suggesting the hybrid architecture navigates the loss landscape more smoothly.

3.3 Quantum Kernel Estimation (QKE)

QKE [1] uses a quantum feature map to define a kernel function for a classical Support Vector Machine.

3.3.1 Quantum Kernel

The quantum feature map $\phi : \mathbb{R}^d \to \mathcal{H}$ maps inputs to a quantum state. The kernel between two inputs is their state overlap:

$$k(\boldsymbol{x}_i, \boldsymbol{x}_j) = |\langle \phi(\boldsymbol{x}_i) | \phi(\boldsymbol{x}_j) \rangle|^2 = |\langle 0 | U_\phi^\dagger(\boldsymbol{x}_i) U_\phi(\boldsymbol{x}_j) | 0 \rangle|^2 \tag{12}$$

The full $N \times N$ kernel matrix is computed, then passed to a classical SVM.

3.3.2 Computational Complexity

QKE requires $\mathcal{O}(N^2)$ quantum circuit evaluations, making it impractical for large datasets ($N > 10^3$) without approximations. For this study, PCA is applied to reduce to $d = n_{\text{qubits}}$ dimensions before QKE.

3.3.3 Theoretical Advantage

The quantum advantage of QKE arises when the quantum feature map produces a kernel that is exponentially hard to compute classically. Whether this advantage materialises for radar features (which are continuous, low-dimensional, and structured) is an empirical question addressed in this paper.

4 Ensemble: VQC + HQNN + GBT

A three-model soft-vote ensemble combines probability outputs:

$$p_c^{\text{ens}} = \frac{1}{3}\left(p_c^{\text{VQC}} + p_c^{\text{HQNN}} + p_c^{\text{GBT}}\right) \tag{13}$$

The rationale is that VQC and HQNN explore the solution space differently from GBT (quantum interference vs. gradient boosting), so their errors may be partially uncorrelated, yielding ensemble gains beyond any individual model.

5 Results

5.1 Overall Performance

Table 1: All models compared — corrected 5-feature set, study dataset.

Model	**Accuracy**	**F1 Macro**	**AUC**	**Notes**
Classical				
Random Forest	89.6%	89.5%	0.989	300 est, balanced
SVM (RBF)	80.8%	80.7%	0.964	$C = 10$, γ =scale
MLP	77.9%	77.6%	0.948	64×64, 55 iter
GBT	90.0%	90.0%	0.987	300 est, depth 5
Quantum ML				
VQC (5 qubits, 3 layers)	72.4%	71.4%	0.901	Best ep: 15
HQNN (5 qubits, 2 layers, h=32)	82.3%	82.8%	0.953	Best ep: 71
Ensemble (soft vote)				
VQC + HQNN + GBT	**84.2%**	84.0%	0.979	203 test samples

Classical models evaluated on 240 test samples; quantum models on 203 test samples (same dataset, same 300-samples/class training pool, slight difference in test fraction between runs). All use corrected 5-feature set with SMOTE and StandardScaler.

5.2 Cargo vs. Tanker: Class-Level Deep Dive

Table 2: Cargo (70) and Tanker (80) recall for all models.

Model	**Type 70 Recall**	**Type 80 Recall**
Baseline (original 5-feat)		
VQC (original)	44%	50%
HQNN (original)	67%	47%
Ensemble (original)	66%	44%
Study dataset (radarfeatureL_Study, 300/class)		
VQC (5 qubits)	66%	44%
HQNN (5 qubits)	**83%**	58%
GBT (300 est)	75%	**78%**
Ensemble (3 models)	81%	69%

5.3 Learning Curves

[Training and validation accuracy vs. epoch plots for VQC and HQNN to be inserted. Key observations:

- VQC typically plateaus early (epoch 8–32) due to barren plateau effects in the gradient landscape.
- HQNN converges more smoothly due to the classical pre/post layers providing richer gradient signal.
- Early stopping (patience=20) prevents overfitting by restoring the best-validation-accuracy checkpoint.]

5.4 Confidence Score Comparison

For each model, the distribution of $\mathrm{conf}(\boldsymbol{x}) = \max_c p_c(\boldsymbol{x})$ is compared between correctly and incorrectly classified samples.

[Violin plots / histograms to be inserted.]

Expected findings based on preliminary results:

- HQNN produces better-separated confidence distributions than VQC.
- GBT tends to overconfidence (many predictions at > 0.95) even on misclassified samples.
- The VQC+HQNN+GBT ensemble produces smoother probability outputs that may offer better calibration.

6 When Does QML Help?

6.1 Theoretical Conditions for Quantum Advantage

Quantum advantage in classification arises under specific conditions:

1. **Kernel inexpressibility**: The quantum kernel captures correlations that classical kernels cannot efficiently compute.
2. **Data structure**: The data lies near a quantum state manifold that angle encoding maps to efficiently.
3. **Small data regime**: Quantum models have fewer parameters (∼75 for VQC vs. thousands for deep classical networks), potentially generalising better from small samples.

6.2 Analysis for the Radar Classification Problem

Table 3: Assessment of quantum advantage conditions for this problem.

Condition	**Satisfied?**	**Evidence**
Low-dimensional structured data ($d \leq 10$)	✓Yes	5 features
Small training set ($N < 10^3$)	Partial	∼1,200–10,000
Feature overlap requiring non-linear decision boundary	✓Yes	70/80 boundary
Quantum kernel advantage	Unknown	Requires QKE evaluation

The small-data regime is where QML most often shows competitive performance relative to classical deep learning. With 5 features and ∼1,200 training samples, HQNN (75 parameters) may generalise better than MLP (thousands of parameters) despite similar capacity. Whether it outperforms GBT (which also has relatively compact parameterisation) is the empirical question.

6.3 Observed QML Behaviour

From training on the study dataset (radarfeatureL_Study, 100,430 observations, 300 samples per class for training):

- **HQNN achieved 82.3% accuracy and 82.8% F1 Macro** with best weights at epoch 71 (early stopping at epoch 91). This outperforms VQC by 10 percentage points and is competitive with RF (89.6%).

- **VQC achieved 72.4% accuracy** (best at epoch 15, early stopping at epoch 35), limited by circuit expressibility and the 5-qubit strongly-entangling ansatz.

- **HQNN converges smoothly over 71 epochs** versus VQC's plateau at epoch 15, confirming the classical encoder/decoder layers provide richer gradient signal than the all-quantum VQC.

- **The VQC+HQNN+GBT ensemble (84.2%)** does not outperform GBT alone (90.0%) on accuracy but achieves competitive AUC (0.979 vs. 0.987) and may offer better-calibrated uncertainty estimates for operational maritime classification systems.

- **Training cost**: VQC and HQNN required approximately 70 minutes combined on CPU using `lightning.qubit` with adjoint differentiation. GBT trained in 1.6 seconds.

7 Discussion

7.1 QML vs. Classical: The Core Trade-off

- **Accuracy**: GBT (90.0%) outperforms all other models. HQNN (82.3%) is the best QML model and closes two-thirds of the gap between VQC and GBT. SVM (80.8%) and MLP (77.9%) fall below HQNN.

- **Training speed**: GBT trains in 1.6 seconds; VQC requires approximately 30 minutes and HQNN approximately 40 minutes on CPU using `lightning.qubit` with adjoint differentiation. On GPU the overhead is higher for 5-qubit circuits (0.24 s per 10 evaluations) than CPU (0.04 s), due to GPU launch overhead dominating at this circuit size.

- **Scalability**: QML simulation cost scales exponentially with qubit count. At 5 qubits, statevector simulation remains tractable and is the regime explored here.

- **Confidence calibration**: HQNN probability outputs via softmax show better-separated confidence between correct and incorrect predictions compared to GBT, which tends to overconfidence (> 0.95) even on misclassified Cargo/Tanker samples.

7.2 Barren Plateau Problem

VQC training can suffer from exponentially vanishing gradients as qubit count grows — the "barren plateau" phenomenon [4]:

$$\mathrm{Var}\left[\frac{\partial L}{\partial \theta_k}\right] \in O(2^{-n}) \tag{14}$$

With $n = 5$ qubits, this is not yet a critical issue, but it explains the earlier plateau of VQC training curves and the benefit of the hybrid HQNN architecture, which mitigates barren plateaus through classical layers.

7.3 Simulator vs. Real Quantum Hardware

All experiments use classical statevector simulation. On real NISQ devices:

- Gate errors and decoherence would degrade performance.
- Shot noise would introduce stochastic gradients.
- Error mitigation techniques would be required.

Demonstrating QML advantage on real hardware for this problem remains future work.

8 Conclusion

This paper provides the first systematic comparison of QML and classical ML for multi-class radar vessel type classification from a coastal X-band surveillance radar. Key findings from training on the radarfeatureL_Study dataset (100,430 rows, corrected 5-feature set):

1. **HQNN (82.3% accuracy, F1 82.8%, AUC 0.953)** is the strongest QML model and approaches the best classical model (GBT, 90.0%) despite having only ∼75 trainable parameters vs. GBT's 300 decision trees.

2. **VQC (72.4%, F1 71.4%, AUC 0.901)** underperforms HQNN by 10 percentage points, limited by the all-quantum circuit's expressibility at 5 qubits and the early barren plateau observed at epoch 15.

3. **GBT (90.0%, AUC 0.987) is the best single model**, benefiting from explicit feature interaction modelling and requiring only 1.6 seconds of training time versus ∼70 minutes for the quantum models.

4. **The VQC+HQNN+GBT soft-vote ensemble (84.2%, AUC 0.979)** does not exceed GBT alone, suggesting the VQC acts as the weaker ensemble member and dilutes the overall prediction.

5. **QML shows measurable advantage over SVM (80.8%) and MLP (77.9%)**: HQNN outperforms both classical neural approaches despite training on an identical feature set.

6. The **Cargo/Tanker boundary** remains the dominant error source for all models: HQNN achieves 83%/58% recall for 70/80, GBT achieves 75%/78%, with the ensemble at 81%/69%.

7. QML advantage on real quantum hardware and larger qubit counts (7–12 qubits) remains an open research question.

References

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